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Networks Project Proposal

Project Name: LoRa and Mesh

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Project Description 3
Research References **3**
Hardware and Software **4**
Specific Subjects We Will Cover **4**
Implementation Plan **5**

Project Description

Our project will explore how LoRa technology allows devices to communicate over long distances without needing Wi-Fi, cellular networks, or Internet access. We will focus on MeshCore, a firmware that allows compatible LoRa devices to exchange messages and forward them through repeaters to extend communication range. We will research how LoRa networks work, compare differe nt firmware options, and build a small network using real devices to demonstrate communication without Internet access.

LoRa is a wireless modulation technique that's made from Chirp Spread Spectrum (CSS) technology. It encodes information on radio waves using chirp pulses. It is something similar to how dolphins or bats communicate with each other.

LoRaWAN is a MAC layer protocol built on top of LoRa modulation. It is a software layer which shows how devices use the LoRa hardware, for example when they transmit, and the format of messages.

MeshCore is a simple, secure, off-grid mesh communications system
You can send messages even when Wifi and Cellular are down. We will be using this network to send and receive messages.

Research References

1. **The Things Network – LoRa and LoRaWAN**
   https://www.thethingsnetwork.org/docs/lorawan/what-is-lorawan/
   Difference between the radio technology and LoRaWAN networking.

2. **MeshCore – Official Website**
   https://meshcore.co.uk/
   Firmware, device setup, companion nodes, repeaters, and other roles.

3. **Meshtastic – Official Documentation**
   https://meshtastic.org/docs/introduction/
   Alternative off-grid LoRa mesh firmware.

4. **Reticulum – Official Documentation**
   https://reticulum.network/manual/
   Reticulum networking and RNode radio interfaces.

Hardware and Software

Hardware:

We already have ESP32 boards that may be useful. We will first check whether they are compatible with LoRa radio modules and MeshCore firmware before deciding. We will also look at other LoRa development boards such as Heltec WiFi LoRa 32 to see if they are suitable for the MeshCore version we will be using. We will most likely need USB cables, antennas, and power supplies or batteries. For the device, it will depend on the MeshCore version.

Software:

Our main system for the demonstration will be MeshCore along with a compatible client to send/recieve messages. We will be using official documentation to configure devices, and we will also research other firmware for comparison such as Meshtastic and Reticulum/RNode. If we decide to use our ESP32s, the Arduino IDE will be useful for testing, but we still have to confirm whether this approach is compatible with our chosen setup.

Specific Subjects We Will Cover

1. **LoRa, LoRaWAN, and MeshCore:** Explain how each technology works, their differences, and how devices can communicate without Internet access.

2. **Real-Life Applications:** Explore how LoRa can be used for emergency communication, agriculture, environmental monitoring, and communication in remote areas.

3. **Firmware:** Compare MeshCore, Meshtastic, and Reticulum/RNode, like their features, differences, and power consumption.

4. **Solar-Powered Repeaters:** Research how to power a LoRa repeater using a solar panel, charge controller, and battery. We will consider Montreal's winter conditions when estimating the required power and battery capacity.

5. **Link Budget and Signal Range:** Explore how transmission power, antenna gain, antenna height, distance, and obstacles such as buildings affect signal strength and communication range.

6. **MeshCore Node Roles:** Explain the different roles in a MeshCore network, including companion nodes, repeaters, room servers, bots, and observers.

Implementation Plan

Our team plans to build a small LoRa network using three compatible devices running MeshCore firmware.

First, we will research and select the hardware needed for our project. Since we already have regular ESP32 boards, we will investigate whether we can use them with compatible LoRa modules or whether we should use development boards with built-in LoRa radios.

Once we have the hardware, we will install MeshCore firmware and configure two devices as companion nodes and one as a repeater.

We will start by testing communication between two devices without an Internet connection. After confirming that messages can be sent and received, we will add the repeater to extend the communication range.

We will also test how distance and obstacles, such as walls and buildings, affect signal strength. If possible, we will demonstrate a message being forwarded through the repeater when direct communication is unavailable.

Finally, we will research the power requirements for a solar-powered repeater that could operate during a Montreal winter. We will estimate the solar panel and battery capacity needed, but we do not plan to build the solar system unless time and resources allow.

During our presentation, we plan to demonstrate the working network using real devices and explain our results.